

VidVaani

Automated, low-cost dubbing of English technical lectures into Hindi

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The problem: technical education is locked in English

- India's best engineering lectures — NPTEL alone hosts **50,000+ hours** — are overwhelmingly in English.
- A large fraction of our students think and learn in Hindi and other Indian languages; a fast English lecture is a barrier, not a resource.
- **NEP 2020** explicitly calls for higher education content in Indian languages.
- Manual dubbing is slow and expensive: NPTEL's human-verified translation effort has taken years to cover a subset of courses in 11 languages.

Goal: given any lecture video, produce a natural-sounding Hindi version in minutes, for rupees — with the technical vocabulary students actually use.

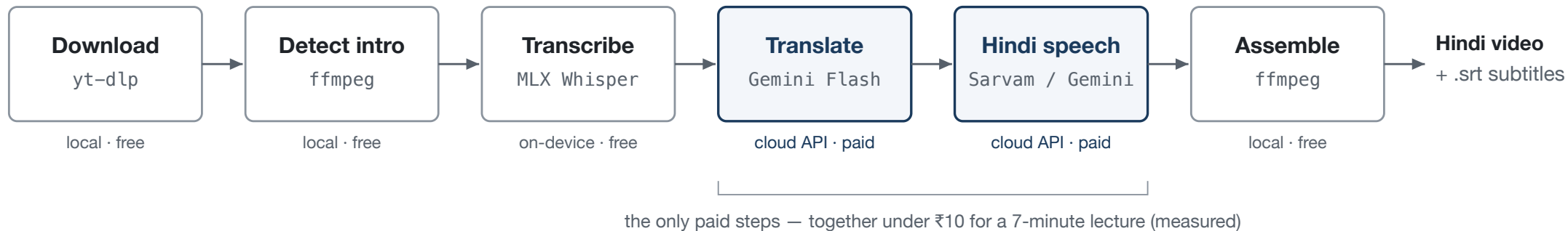
NPTEL translation initiative: nptel.ac.in/translation · Bhashini (MeitY) has dubbed ~200 NPTEL/SWAYAM courses in 8 languages via human-in-the-loop workflows.

Existing options do not fit the lecture use case

Option	Cost	Why it falls short
YouTube auto-dubbing	Free	Only the channel owner can enable it; dubs are uneditable; no control over voice or terminology; widely criticised as robotic
Commercial dubbing SaaS (ElevenLabs, Rask, HeyGen, Dubverse, Murf)	₹3,000–11,000 per lecture	Priced for short marketing clips; monthly minute caps; each extra language billed again
Azure Video Translation	≈ ₹2,000 per lecture	Cheapest managed option, but limited voice choice and no terminology control
NPTEL / Bhaskar Video	Commercial	Highly limited control over content

VidVaani: a six-stage open pipeline

Input: YouTube URL

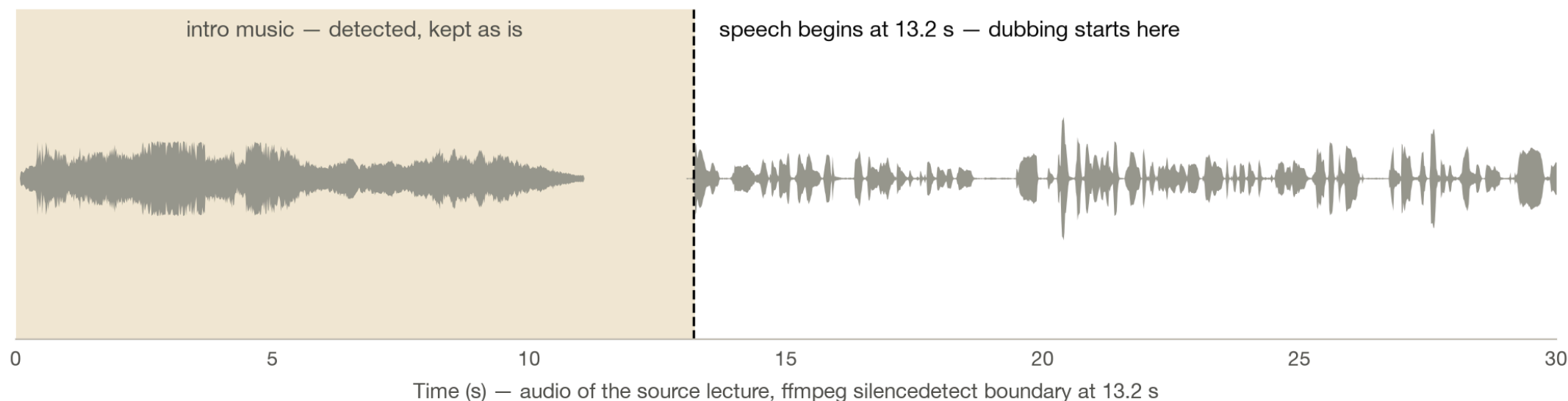


- Single command: `vidvaani dub <URL> --full -b sarvam -v abhilash`
- Transcription runs **on-device** (MLX Whisper on Apple Silicon) — audio never leaves the machine until translation.
- Every intermediate artifact is cached: re-dubbing with a new voice skips download, transcription, and translation.

The next five slides walk **one real lecture** — NPTEL CS7015 Lecture 1.1 (Prof. Mitesh

Steps 1–2 — Download, and find where speech starts

`yt-dlp` fetches any YouTube URL or local file — no allowlist, no channel ownership needed. Then `ffmpeg silencedetect` locates the boundary between intro music and speech:



The music is **not dubbed over** — it is kept exactly as published, and dubbing begins on the first spoken word.

Step 3 — Transcribe on-device

MLX Whisper (`distil-large-v3`) runs locally on Apple Silicon — the audio never leaves the machine. Output: time-stamped English segments, grouped to ≤ 15 s:

Time	Whisper output (verbatim)
18.6 – 29.9 s	"In today's lecture is going to be a bit non-technical. We are not going to cover any technical concepts..."
29.9 – 41.1 s	"So, we hear the terms artificial neural networks, artificial neurons quite often these days..."
41.1 – 56.0 s	"And this history contains several spans across several fields, not just computer science..."

Transcribing the full 7-minute lecture took **17–20 s** (measured) — and it is free.

Revs taken verbatim from the pipeline transcript file (`4TQ5c_vNK6c_transcript_en.json`)

Step 4 — Translate for the clock, and the classroom

Constraints in the translation prompt: *speakable in the same duration; conversational (not literary) Hindi; technical terms stay in English.*

	Original (English)	VidVaani (Hindi)
13.2 s	"Hello everyone, welcome to lecture 1 of CS7015 which is the course on deep learning ."	"सभी को नमस्कार, CS7015 के लेक्चर एक में आपका स्वागत है, जो डीप लर्निंग का कोर्स है।"
29.9 s	"...we hear the terms artificial neural networks , artificial neurons quite often these days."	"...आजकल हम आर्टिफिशियल न्यूरल नेटवर्क्स, आर्टिफिशियल न्यूरॉन्स शब्द अक्सर सुनते हैं।"

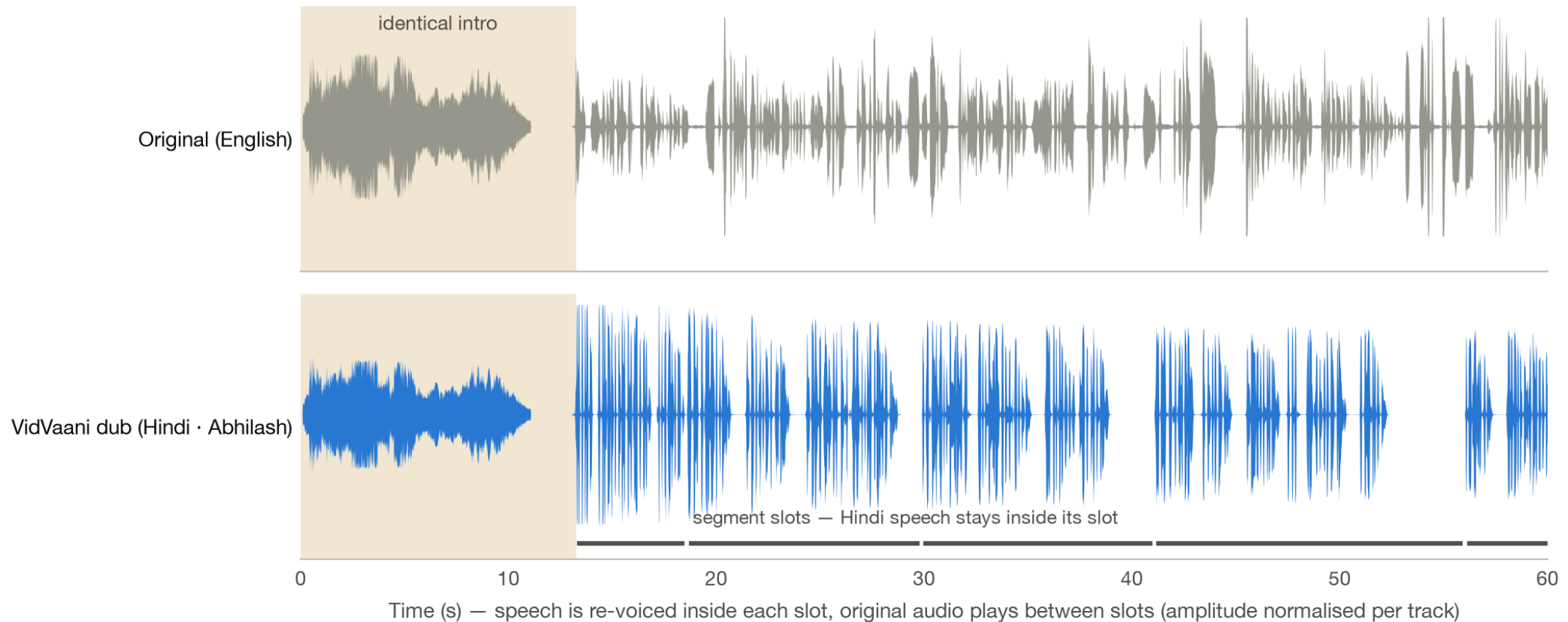
Students say *gradient* and *neural network* — a dub that renders them as प्रवणता and तंत्रिका जाल (as fully-literal systems do) sounds foreign to the very audience it serves.

Step 5 — Synthesize, and fit the clock

Hindi runs ~15–35% longer than English. Per segment, following the automatic-dubbing literature:

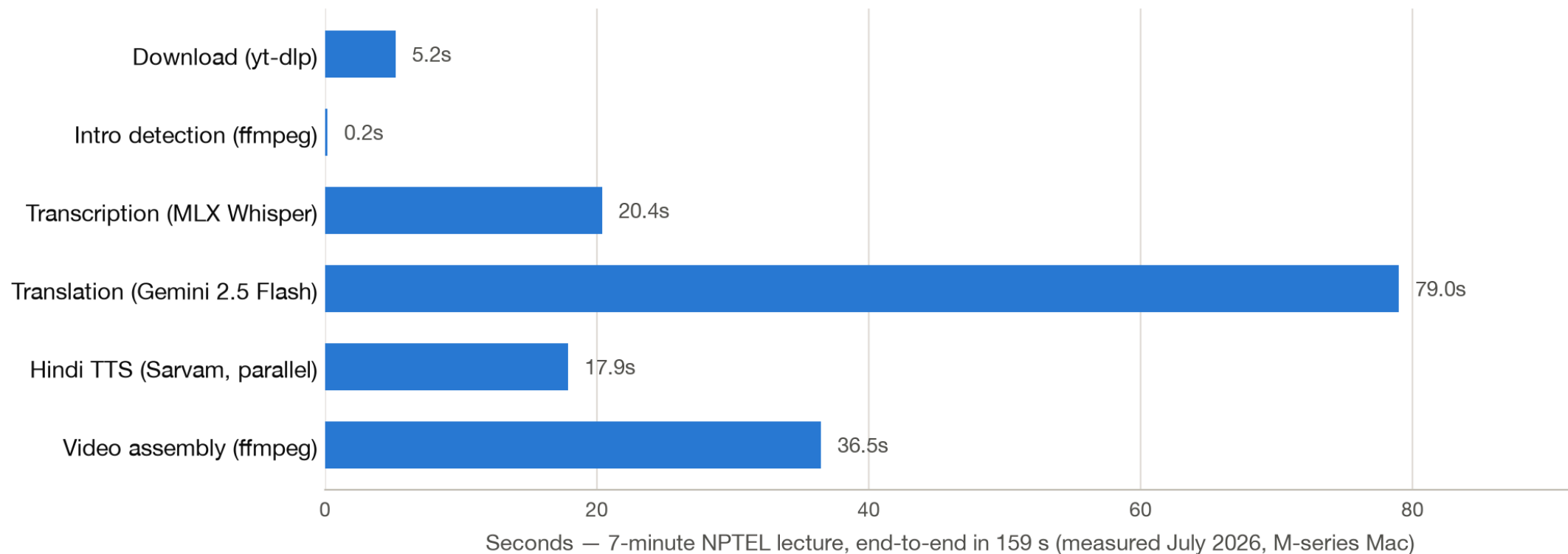
1. The translator gets a **word budget** per slot (2.4 words/s, calibrated against the TTS voice's measured 2.8 words/s delivery) — fixing length at translation time beats stretching audio afterwards.
2. Synthesize at natural pace, trim trailing silence, measure with `ffprobe`.
3. If it fits ($\pm 5\%$) — done. Otherwise speed-adjust with an **asymmetric clamp** ($0.95\times$ – $1.35\times$): listeners tolerate faster speech far better than slowed speech.
4. Still too long? The clip **spills into the trailing pause** rather than being cut — professional dubs sacrifice timing, never content.
5. During pauses, the **original soundtrack plays** — atmosphere and intro music survive the dub.

Step 6 — Reassemble: the result, seen in the signal



The intro is bit-identical to the source; every Hindi utterance starts on its original segment boundary; between utterances the original soundtrack plays. Plus a Hindi `.srt`, and

Measured performance — faster than real time



A **7-minute lecture dubs in 2½–4 minutes** end-to-end on a laptop — the bottleneck is the translation API, not synthesis. Re-runs with a different voice take under a minute (cached transcript + translation).

Measured cost — rupees, not lakhs

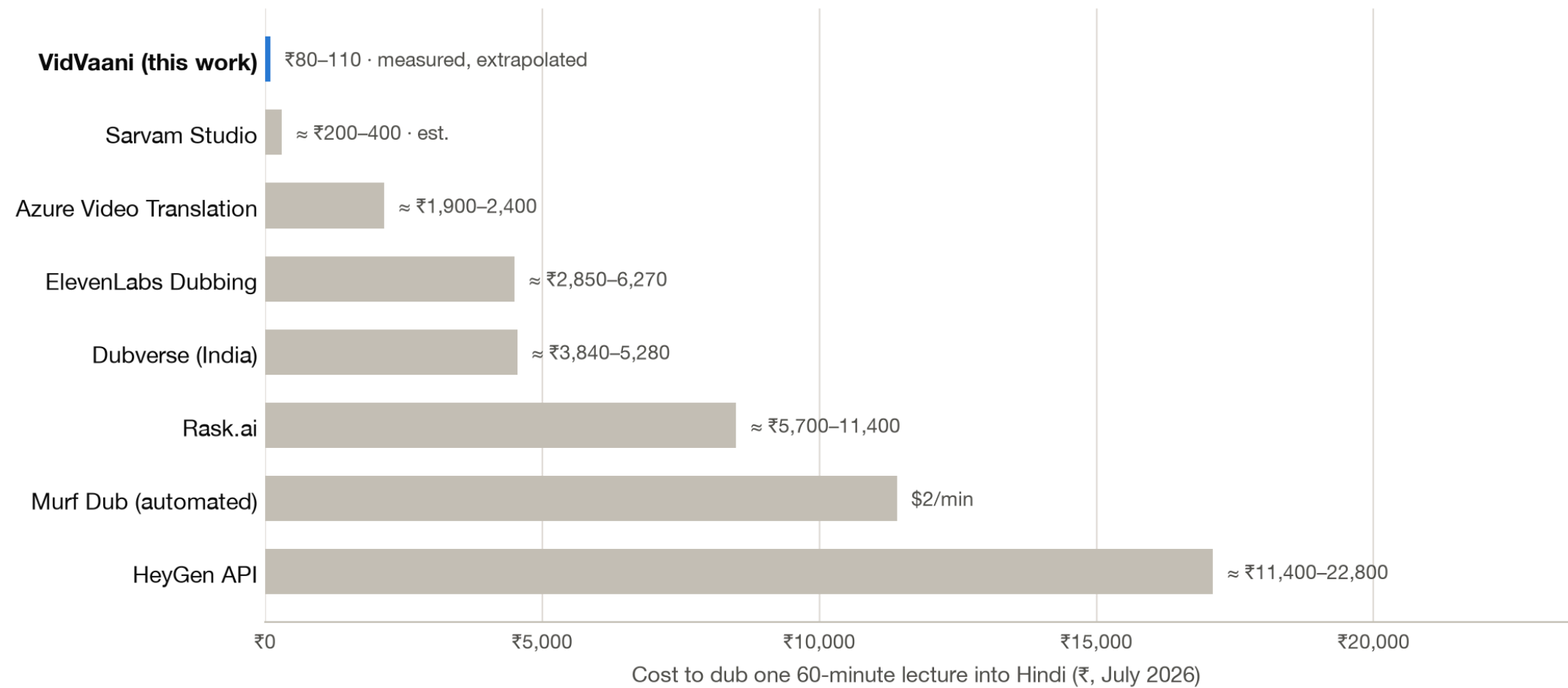
Measured on the same 7-minute lecture (5,307 Hindi characters):

	Translation	Hindi TTS	Total (7 min)	Extrapolated, 1 hour
Sarvam <code>bulbul:v2</code>	₹1.4	₹8.0	₹9.4 (\$0.10)	≈ ₹80
Gemini 2.5 Flash TTS	₹1.4	₹9.6	₹11.0 (\$0.12)	≈ ₹95
Edge TTS (free tier)	₹1.4	₹0	₹1.4	≈ ₹12

- A full **40-lecture course** ≈ ₹3,200–4,400 — less than one SaaS-dubbed lecture.
- Costs measured by the pipeline itself from API-reported token counts, at official July 2026 prices.

Stable across runs: after calibrating the translation word budget, the same lecture re

Where this sits in the market



Midpoints of published prices, single target language, July 2026. YouTube auto-dubbing is

Why not just use the platforms?

YouTube auto-dubbing

- Now rolled out to all channels — **but only the uploader** can enable it.
- A university cannot dub NPTEL or MIT content it does not own.
- Dubs cannot be edited — a mistranslated technical term can only be unpublished, not fixed.

Sarvam Studio & SaaS dubbers

- Produce formal, fully-translated Hindi (क्षेत्रफल for "area") — jarring for STEM learners used to Hinglish.
- No control over segment-level phrasing or terminology.
- VidVaani uses the **same Sarvam voices** with full editorial control, at API prices.

What a run looks like

```
$ vidvaani dub "https://youtube.com/watch?v=4TC5s_xNKSs" --full -b sarvam -v anushka

Downloaded: Deep Learning(CS7015): Lec 1.1 Biological Neuron
Transcribed: 33 segments                               Generated subtitles: 4TC5s_xNKSs_hindi.srt
TTS: 33/33 segments (parallel)                         Created: 4TC5s_xNKSs_hindi_anushka.mp4
```

Timing Breakdown			API Costs		
Download	5.2s	3.3%	Translation	9,542 tokens	\$0.015
Transcription	20.4s	12.8%	TTS (Sarovam)	5,307 chars	Rs 7.96
Translation	79.0s	49.6%	Total		Rs 9.4
TTS Generation	17.9s	11.2%			
Video Assembly	36.5s	22.9%			
Total	159.2s	100%			

Every run reports its own timing and cost — the numbers on the previous slides are this output, not estimates.

Demonstrations

nipunbatra.github.io/vidvaani — plays in any browser:

1. **58-second calculus clip, eight ways** — original, Sarvam and Gemini voices (same translation, so the comparison is purely the voice), plus Sarvam's own Dashboard output.
2. **Full 7-minute NPTEL lecture** — original beside Hindi dub; intro music preserved; five voices; burned-in subtitles.

What to listen for:

- Technical terms stay in English (*deep learning, CS7015*).
- Pauses land where the professor pauses.
- Native Indian prosody (Sarvam) vs. slight Western accent (Gemini).

Honest limitations

- **No human review loop yet** — NPTEL's own effort treats faculty verification as essential for pedagogy; ₹80/lecture buys the draft, not the sign-off.
- **Timing fit is still partly post-hoc** — word budgets plus a 0.95–1.35× speed clamp handle most segments, but very dense passages can still sound hurried (see next slide).
- Gemini TTS models are **preview** — pricing and behaviour may change; Edge TTS is an unofficial free endpoint, fine for prototyping only.
- No lip-sync. Voice cloning of the original lecturer now works locally (next slide) and stays gated on **consent** — the one published cloned demo is the author's own voice on his own lecture.
- Hindi first; Sarvam voices cover 11 Indian languages, so extension is natural.

Towards a fully local pipeline

Every cloud stage now has a working local, open-weights replacement — measured on the 58 s demo clip, Apple Silicon, July 2026:

Stage	Cloud (current)	Local (experimental)	Measured, local
Transcribe	— (already local)	MLX Whisper	17–20 s per 7-min lecture
Translate	Gemini 2.5 Flash	Gemma 4 31B (MLX 4-bit, Apache-2.0)	59 s for the clip (12B via ollama: 25 s)
Hindi speech	Sarvam API	Qwen3-TTS 1.7B (MLX, Apache-2.0)	1.6× real time , cloned (first run: ~4×)

- The "Fully local" demo card was produced this way — **₹0 per lecture, nothing leaves**

Cloning the lecturer's voice: five rungs in one day

Similarity = ECAPA cosine to the lecturer's real voice. His own recordings score **0.93** against each other (the ceiling); other voices score **~0.3**. All measured 12 Jul 2026.

Rung	Method	Compute	Dub similarity
1	Single take, ICL cloning from a 28 s reference	laptop, 95 s	0.76
2	Best-of-16 search (4 curated refs × temp × seed), scored + stitched	laptop, ~45 min	0.85
3	+ LoRA fine-tune, rank 16 on 30 min of his verified speech	laptop (MLX), 7.5 min training	0.86
4	+ official full SET community, corrected L P	lab A100, ~3	0.89

Roadmap: smarter time alignment

We surveyed the automatic-dubbing literature (Amazon, Microsoft, IIT Madras, IWSLT) and re-based the alignment design on it — full notes in `docs/timing-alignment.md` :

Already implemented — word budgets in the translation prompt; asymmetric speed clamp; spill-into-pause instead of truncation; trailing-silence trimming.

Next:

1. **TTS-native pace instead of waveform stretching** — Sarvam exposes `pace` (0.3–3.0×); synthesis at the right speed beats resampling in every published comparison.
2. **Re-translate outliers once** — segments still needing $> 1.3\times$ speed get one "shorten by 25%" LLM pass; $\sim 25\%$ of En $\rightarrow$ Hi segments are expected to qualify.
3. **Sentence-boundary segmentation** with word-level timestamps ($\approx +4.5$ BLEU over blind packing).

Summary

- **Any lecture → natural Hindi dub in minutes:** one command, six automated stages, technical vocabulary preserved.
- **Measured, not estimated:** a 7-min lecture in ~3 minutes for under ₹10; ≈ ₹80–110 per lecture-hour; a 40-lecture course for the price of a textbook.
- **20–100× cheaper** than commercial dubbing platforms, with editorial control none of them offer.
- Open source (MIT): github.com/nipunbatra/vidvaani
- Live demos: nipunbatra.github.io/vidvaani

Next: faculty-in-the-loop review, more Indian languages, and a student comprehension study.